4.A Chemical Reactions and Aqueous Solutions I

Chemical Equations. Chemical reactions involve the transformation of one set of chemical substances (reactants) into another (products). Bonds between atoms are made and broken and atoms change places as a result. Chemical equations are symbolic representations of chemical reactions; they provide both macroscopic and submicroscopic information and work at either scale.

- **Chemical equations** represent the transformation of one or more chemical substances into new substances.
- **Reactants (starting materials)** are the initial components of a chemical reaction.
- **Products** are the components at the end of the reaction.
- Chemical equations are always written…

- A wealth of information can be extracted from chemical equations.
    - **Stoichiometric coefficients** describe the numbers of each type of molecule or formula unit consumed or produced during the process.
    - **Phase designators** indicate the state of matter of each substance at the macroscopic scale.

	Notation	Example
What happens?	Chemical formulas of the reactants and products	$C_2H_4 + Cl_2 \rightarrow C_2H_4Cl_2$
In what proportions?	Coefficients before each chemical formula	$\mathbf{2}\ H_2 + O_2 \rightarrow \mathbf{2}\ H_2O$
In what physical states?	(*s*), (*l*), (*g*) after each chemical formula	$\mathbf{2}\ H_2(g) + O_2(g) \rightarrow \mathbf{2}\ H_2O(l)$
Under what conditions?	Special conditions written above or below the arrow	$2\,H_2\,(g) + O_2\,(g) \xrightarrow{\text{heat}} H_2O\,(l)$

- In a **balanced** chemical equation, the numbers of atoms of each element on the reactant side are equal to the numbers of atoms of each element on the product side. All physically valid chemical equations must be balanced (conservation of mass).
    - $SO_2 + O_2 \rightarrow SO_3$ is unbalanced.
    - $2\ SO_2 + O_2 \rightarrow 2\ SO_3$ is balanced.

- You should be able to rapidly balance chemical equations and easily verify that an equation is balanced.

- Stoichiometric coefficients are adjusted to balance chemical equations.
  - Do not modify subscripts in chemical formulas; that changes chemical identity. (Remember, compounds are defined by their formulas.)
  - Typically, we strive for the equation with the lowest whole-number coefficients possible, although fractional coefficients are completely fine.

- The process below can be used to balance an equation systematically.

  1. Balance unchanging polyatomic ions as units if they are present in both the reactants and products.
  2. Balance elements that appear in only one reactant and one product.
  3. Balance any remaining elements; resolve odd/even issues by doubling previously determined coefficients.
  4. Verify that the number of each type of atom is the same in the reactants and products and that the coefficients are in the simplest form.

4.1. Ammonia (NH_3) reacts with oxygen gas to form water and nitrogen dioxide.

Write chemical formulas for the reactants and products. What is the coefficient for oxygen when this reaction is balanced with the smallest possible whole-number coefficients? Enter your response as a numeral.

4.2. What is the coefficient for each chemical species when the following reaction is balanced with the smallest possible whole-number coefficients? To answer this question, enter the *sum* of all the coefficients.

$$\_\_\ Al_2(CO_3)_3 + \_\_\ H_3PO_4 \longrightarrow \_\_\ AlPO_4 + \_\_\ CO_2 + \_\_\ H_2O$$

Classifying Chemical Reactions. Recognizing patterns in chemical reactivity helps us simplify our knowledge base and draw analogies between related reactions. Here, we introduce a few simple classes of chemical reactions. Although these classes are not exhaustive, they do cover a lot of ground, and will serve as a starting point for analyzing chemical reactions in the future.

- **Synthesis reactions** involve the combination of simple reactants, often elements, to form a single more complex product.

$$4\ Fe(s) + 3\ O_2(g) \rightarrow$$

- **Decomposition reactions** involve the breakdown of a complex reactant into simpler products, often elements.

$$2\ H_2O(l) \rightarrow$$

- **Single-replacement (single-displacement) reactions** occur when an element reacts with a compound and displaces one of the elements in that compound, producing a new compound and a new element.

$$Zn(s) + 2\ HCl(aq) \rightarrow$$

- **Double-replacement (double-displacement) reactions** occur when two aqueous ionic compounds exchange ions, forming two new compounds.
  - In a **precipitation reaction**, one of the product is an ionic solid.

$$2\ KI(aq) + Pb(NO_3)_2(aq) \rightarrow$$

  - **Acid-base reactions** are double-replacement reactions in which one of the "ions" exchanged is H^+.
    - The **acid** is the reactant containing the H^+ transferred; the transferred H is often written first in the formula.
    - The **base** is the reactant that accepts or gains H^+; bases often contain the OH^- anion.

$$HCl(aq) + NaOH(aq) \rightarrow$$

$$H_2SO_4(aq) + 2\ KOH(aq) \rightarrow$$

- **Combustion reactions** involve the rapid combination of a substance, often a hydrocarbon C_xH_y, with oxygen (O_2) to form carbon dioxide (CO_2) and water (H_2O).

$$C_3H_8(g) + 5\ O_2(g) \rightarrow$$

$$2\ C_{24}H_{50}(s) + 73\ O_2(g) \rightarrow$$

- Summary of reaction types

Reaction Type	General Formula	Examples
Synthesis	$A + B \rightarrow AB$	$4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s)$
Decomposition	$AB \rightarrow A + B$	$2\ H_2O(l) \rightarrow 2\ H_2(g) + O_2(g)$ $2\ KClO_3(s) \rightarrow 2\ KCl(s) + 3\ O_2(g)$
Single replacement	$A + BC \rightarrow AC + B$	$Zn(s) + 2\ HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$
Double replacement	$AB + CD \rightarrow AD + CB$	$2\ KI(aq) + Pb(NO_3)_2(aq) \rightarrow PbI_2(s) + 2\ KNO_3(aq)$ $HBr(aq) + KOH(aq) \rightarrow KBr(aq) + H_2O(l)$
Combustion	$C_xH_y + O_2 \rightarrow CO_2 + H_2O$	$C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(l)$

4.3. Label each chemical reaction as *synthesis* (1), *decomposition* (2), or *single replacement* (3).

$$2\ HgO(s) \rightarrow 2\ Hg(l) + O_2(g)$$

$$2\ NO(g) + O_2(g) \rightarrow 2\ NO_2(g)$$

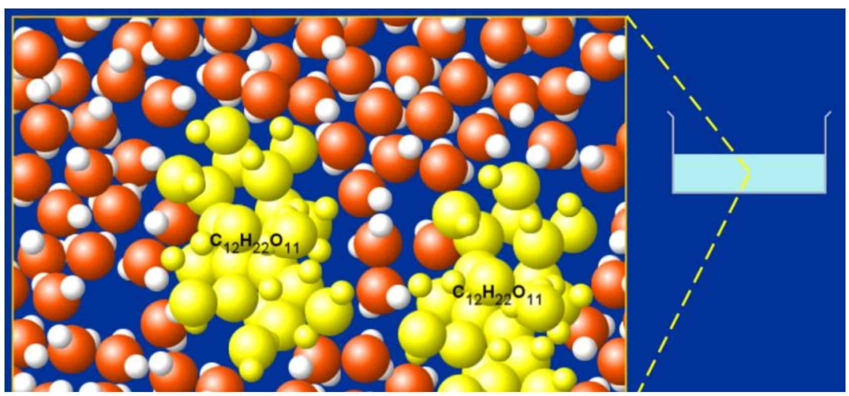
$$2\ Na(s) + 2\ H_2O(l) \rightarrow 2\ NaOH(aq) + H_2(g)$$

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

4.B Chemical Reactions and Aqueous Solutions II

Aqueous Solutions and Solubility. A solution is a homogeneous mixture of two or more components. Liquid solutions in which water is the solvent are called aqueous solutions…and they are *everywhere*.

- Solutions consist of one major component, the **solvent**, and one or more minor components called **solutes**.
- In all solutions, we say that the solutes dissolve in (or are dissolved in) the solvent.
- Liquid solutions in which water is the solvent are called **aqueous solutions**.
  - Solutes in aqueous solution with chemical formula X are denoted "X(aq)."
  - "X(aq)" is also sometimes loosely used to refer to the aqueous solution of X as a whole.



Click above to access interactive simulations of aqueous solutions.

4.4. Rubbing alcohol is a solution that contains approximately 180 grams of isopropyl alcohol (C_3H_8O) for every 100 grams of water (H_2O). Identify the solute and solvent in this solution. How do you know?

Hint: masses can be misleading. What is the *number* ratio of C_3H_8O to H_2O in this solution?

- The process of a compound dissolving into a solvent is called **dissolution**. Solute molecules are surrounded by solvent molecules and separate from one another. The solute “disappears into” the solvent.
    - Unreactive covalent compounds such as alcohols and sugars dissolve in water without any further change. For example, dissolution of the solid sugar sucrose ($C_{12}H_{22}O_{11}$) in water is written as follows.

$$C_{12}H_{22}O_{11}(s) \rightarrow C_{12}H_{22}O_{11}(aq)$$

    - When an ionic compound dissolves in water, the compound **dissociates** as each ion is surrounded by water molecules. This process is called **dissociation**; we generally assume it to be complete.

$$NaCl(s) \rightarrow Na^{+}(aq) + Cl^{-}(aq)$$

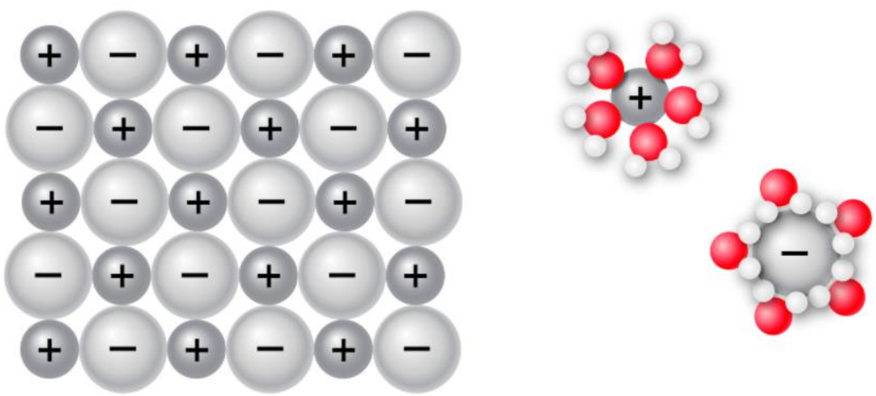


Figure 4.14. Sodium (+) and chloride (–) ions are surrounded by water molecules when sodium chloride is dissolved in water. The surrounded ions are said to be **hydrated**.

4.5. Which statement is *true* regarding dissociation and dissolution?

A. Dissociation can occur without dissolution.
B. Dissolution can occur without dissociation.
C. Dissolution and dissociation can only occur together.
D. None of these statements is true.

- We are often interested in whether a given solute can dissolve in a given solvent.
    - **Soluble** compounds can dissolve in the given solvent to a large extent.
    - **Insoluble** compounds cannot dissolve in the given solvent.

- In CHEM 1212K, we will learn that solubility is a continuum. Throughout CHEM 1211K, we will treat solubility as a binary property. (It often is, essentially.)

- The **solubility rules** are a set of guidelines used to determine whether a given ionic salt (with known cation and anion) is soluble or insoluble in water.

Electrolyte Solutions. Aqueous solutions that conduct electricity are used in a variety of electrical applications, such as redox flow batteries. Whether a given solution does or does not conduct electricity can be inferred from the chemical identity of the solute(s). We'll learn how to make that inference here.

- A solution that contains dissolved ions conducts electricity via motion of these ions; such solutions are called **electrolytes**.

    - **Strong electrolytes** contain a high concentration of dissolved ions and exhibit high conductivity and low resistance to current flow.

    - **Weak electrolytes** contain a low concentration of dissolved ions and exhibit low but measurable conductivity.

    - **Nonelectrolytes** contain essentially no dissolved ions and exhibit zero conductivity.

- Strong electrolytes include aqueous solutions of…
    - Soluble ionic compounds

    - **Strong acids:** compounds HX that, when dissolved, react *completely* with water to form H_3O^+(aq) and X^-(aq).

- Weak electrolytes include aqueous solutions of…
    - **Weak acids:** compounds HA that, when dissolved in water, react *partially* with water to form only small amounts of H^+(aq) and A^-(aq).

- **Weak bases:** compounds B that, when dissolved in water, react *partially* with water to form only small amounts of OH^-(aq) and HB^+(aq).

- **Strong bases** are soluble hydroxide salts; these form strong electrolytes when dissolved in water. All other bases are weak, including insoluble hydroxide salts and amine bases.

- We will concern ourselves with seven (and *only* seven) **strong acids**.
  All other acids are weak…for our purposes.
  - For H_2SO_4, only the first proton transfer is complete. HSO_4^- is not a strong acid.
  - $HClO_3$ is considered weak in some sources.

Compound	Formula	Ions
Lithium hydroxide	LiOH	Li^+, OH^-
Sodium hydroxide	NaOH	Na^+, OH^-
Potassium hydroxide	KOH	K^+, OH^-
Calcium hydroxide	$Ca(OH)_2$	Ca^{2+}, OH^-
Strontium hydroxide	$Sr(OH)_2$	Sr^{2+}, OH^-
Barium hydroxide	$Ba(OH)_2$	Ba^{2+}, OH^-

Compound	Formula	Ions
Hydrochloric acid	HBr	H^+, Br^-
Hydrobromic acid	HCl	H^+, Cl^-
Hydroiodic acid	HI	H^+, I^-
Nitric acid	HNO_3	H^+, NO_3^-
Sulfuric acid	H_2SO_4	H^+, SO_4^{2-}
Perchloric acid	$HClO_4$	H^+, ClO_4^-
Chloric acid	$HClO_3$	H^+, ClO_3^-

- Figure 4.17 gives you an opportunity to explore different compounds to determine whether they are strong electrolytes, weak electrolytes, or nonelectrolytes.

Solution Type	Compound Type	Examples
Strong electrolyte	Soluble ionic salts	NaCl(aq), K_2SO_4(aq)
	Strong bases	NaOH(aq), KOH(aq)
	Strong acids	HCl(aq), HNO_3(aq)
Weak electrolyte	Weak acids	HNO_2(aq), H_3PO_4(aq)
	Weak bases	NH_3(aq), CH_3NH_2(aq)
Nonelectrolyte	Molecular and non-acidic	$C_6H_{12}O_6$(aq) and other sugars

4.6. Which of the following solutes produce an electrolyte solution (strong or weak) when dissolved in water? Select all that apply.

KOH

Sodium phosphate

Sucrose ($C_{12}H_{22}O_{11}$)

Copper(II) chloride

Hydrobromic acid

HNO_2

CH_3CH_2COOH

$HClO_4$

4.C Chemical Reactions and Aqueous Solutions III

4.7. Which of the following are considered weak acids? Select all that apply.

Nitric acid

HF

CH_4

Sodium hydroxide

H_2SO_4

Acetic acid

The Solubility Rules. We previously distinguished between soluble and insoluble compounds. For ionic solids combined with solvent water, several patterns in solubility have been observed. That is, certain cations and anions tend to find themselves in either soluble or insoluble salts. These patterns are colloquially known as the "solubility rules" and we will soon use them to predict the outcomes of chemical reactions.

- The **solubility rules** are a set of guidelines that enable us to efficiently determine whether a given ionic salt is soluble or insoluble. Most guidelines include exceptions.

Soluble ionic compounds contain these ions…	Exceptions
Group 1 cations: Li^+, Na^+, K^+, Rb^+, Cs^+; NH_4^+	—
$C_2H_3O_2^-$, HCO_3^-, NO_3^-, ClO_3^-, ClO_4^-	—
Cl^-, Br^-, I^-	Salts of Ag^+, Hg_2^{2+}, and Pb^{2+}
F^-	Salts of group 2 cations, Pb^{2+}, and Fe^{3+}
SO_4^{2-}	Salts of Ag^+, Ba^{2+}, Ca^{2+}, Hg_2^{2+}, Pb^{2+}, and Sr^{2+}

Insoluble ionic compounds contain these ions…	Exceptions
CO_3^{2-}, SO_3^{2-}, CrO_4^{2-}, PO_4^{3-}	Salts of group 1 cations and NH_4^+
S^{2-}, OH^-	Salts of group 1 cations and Ba^{2+}

4.8. Which statement is *true* regarding an aqueous solution of barium hydroxide? Note that barium hydroxide is a soluble salt.

A. The solution contains many intact $Ba(OH)_2$ formula units.
B. The solution contains equal amounts of barium ions and hydroxide ions and very few (if any) $Ba(OH)_2$ formula units.
C. The solution contains equal amounts of barium ions, hydroxide ions, and $Ba(OH)_2$ formula units.
D. The solution contains equal amounts of barium ions and $Ba(OH)_2$ formula units. It contains twice as many hydroxide ions as barium ions and $Ba(OH)_2$ formula units.
E. The solution contains twice as many hydroxide ions as barium ions, and it contains very few $Ba(OH)_2$ formula units.

Precipitation Reactions. When two aqueous solutions of ionic salts are mixed, a cation and anion that constitute an insoluble salt may suddenly find themselves in contact. In this case, a reaction occurs as the ions quickly organize themselves into a solid and “crash out” of solution. Here, we’ll learn how to apply the solubility rules to predict the outcome when ionic solutions are mixed like this. Sometimes, nothing happens at all!

- When two aqueous solutions of ionic compounds are mixed, ions may come into contact that can form an insoluble ionic salt. The result is a **precipitation** reaction.

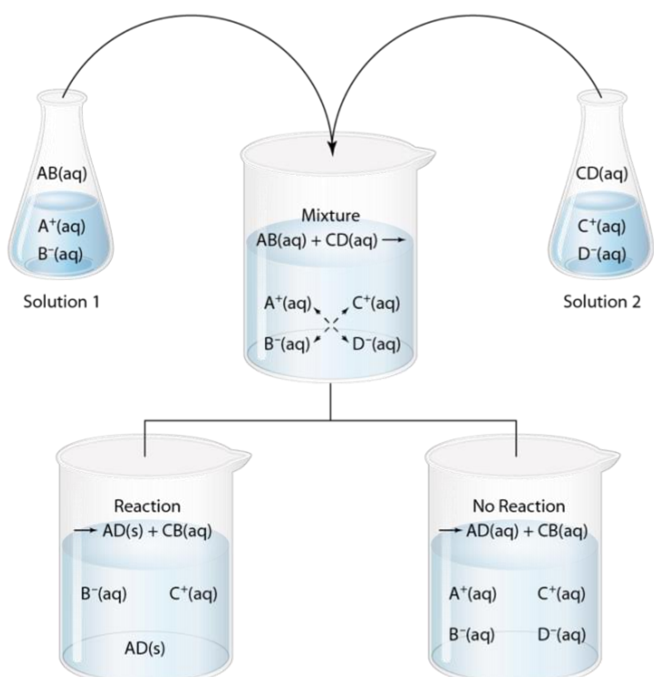


Figure 4.19. Mixing two aqueous solutions may result in formation of a **precipitate** AD (left) or no reaction (right).

- When considering two ionic salts in solution together, we "switch the dance partners" and look for an insoluble salt using the solubility rules.
    - If none exists, no reaction occurs.
    - An insoluble salt that does form is called a **precipitate**. The two ions remaining in solution didn't undergo any chemical change; we call them **spectator ions**.

Example. Write a balanced chemical equation for the reaction of aqueous potassium sulfate with aqueous barium nitrate.

4.9. What is the *name* of the insoluble product formed when potassium carbonate reacts with iron(III) carbonate?

Representing Ionic Reactions. Ionic salts are always neutral, and so cations are always accompanied by anions (and *vice versa*). However, we've seen that for precipitation reactions (and other reactions of ions), there are often extra ions that don't react, but just follow the reactive ions around to balance charge. It's convenient and efficient to leave these spectator ions out of chemical equations. We'll learn how to do that here. We like net ionic equations because they represent the thermodynamic essences of ionic reactions!

- A **total equation** or **molecular equation** lists complete chemical formulas for dissolved ionic compounds, without consideration of dissociation in aqueous solution.

$$3\ K_2CO_3(aq) + 2\ FeCl_3(aq) \rightarrow Fe_2CO_3(s) + 6\ KCl(aq)$$

- A **complete ionic equation** shows each *aqueous* ion as a separate species with its charge. Spectator ions appear on both the reactant and product sides.

- Subtracting spectator ions from both sides gives the **net ionic equation**, which focuses on species undergoing chemical change.

Example. Write the total ionic equation for the reaction that occurs when aqueous solutions of aluminum sulfate and barium chloride are mixed.

4.10. Identify the number and type of spectator ions in the reaction of aqueous solutions of aluminum sulfate and barium chloride. Which of the following are spectator ions?

A. Ba^{2+}
B. Cl^{-}
C. Al^{3+}
D. SO_4^{2-}

Write the net ionic equation for the reaction that occurs when aqueous solutions of aluminum sulfate and barium chloride are mixed.

Chapter Summary. Some of the amazing things we can now do with chemical reactions and aqueous solutions…

1. Interpret chemical equations to infer the number ratios in which substances are consumed and produced, the states of matter of the reactants and products, and the conditions of the reaction.

2. Balance chemical equations.

3. Classify chemical reactions as synthesis, decomposition, single replacement, double replacement, or combustion processes.

4. Describe and visualize solutions, particularly aqueous solutions of molecular (covalent) and ionic solutes; distinguish between dissolution and dissociation.

5. Identify a given aqueous solution as a strong, weak, or non-electrolyte.

6. Apply the solubility rules to infer whether a given ionic salt is soluble or insoluble in water.

7. Predict the outcome with respect to precipitation (or not) when aqueous solutions of ionic salts are mixed with one another.

8. Write molecular, total ionic, and net ionic equations for precipitation reactions.